SIMATS ENGINEERING
ASSESSMENT TOOL 1-APPLICATION- BASED QUESTIONS

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5103

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:100**

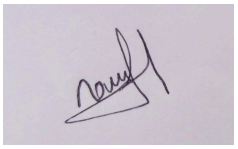
Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I Naveen G / 192411019 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



APPLICATION- BASED QUESTIONS

Scenario: Event Communication Security A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.
Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

Scenario: Intercepted Suspicious Message A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.
Deduce the most likely original message.
Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

Answer

Scenario 1: Event Communication Security

A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message **"MEET AT NOON"** is transformed into an encoded version using a fixed shift pattern.

Question:
Reconstruct how the encrypted message might appear if a shift of 4 positions is applied. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

Answer:

The given plaintext message is:

MEET AT NOON

A **Caesar Cipher** with a shift of **4 positions** is used. In this method, each letter is replaced by the letter that occurs four positions after it in the alphabet.

For example:

- M → Q
- E → I
- T → X
- A → E
- N → R
- O → S

Therefore:

Plaintext:
MEET AT NOON

Ciphertext:
QIIX EX RSSR

The spaces between the words are preserved.

To decrypt the message, an authorized member who knows that the encryption uses a shift of 4 moves each letter **four positions backward** in the alphabet.

For example:

- Q → M
- I → E
- X → T
- E → A
- R → N
- S → O

Thus:

QIIX EX RSSR → MEET AT NOON

Hence, the Caesar Cipher uses the same secret shift value in both directions: **+4 for encryption and −4 for decryption**.

Scenario 2: Intercepted Suspicious Message

A cybersecurity intern intercepts the coded message:

“GSRH RH Z HVXIVG”

during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

Question:
Deduce the most likely original message. Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

Answer:

The most likely original message is:

THIS IS A SECRET

The ciphertext is generated using the **Atbash Cipher**, which is a simple substitution cipher. In the Atbash cipher, the alphabet is reversed, so that:

- A ↔ Z
- B ↔ Y
- C ↔ X
- D ↔ W
- …
- H ↔ S
- I ↔ R
- S ↔ H
- T ↔ G

Applying this substitution to the given ciphertext:

GSRH RH Z HVXIVG

gives:

THIS IS A SECRET

The pattern of the words also supports this deduction. The four-letter word **GSRH** can correspond to the common word **THIS**. The repeated two-letter word **RH** appears twice, matching the repeated word **IS**. The single-letter word **Z** corresponds to **A**. Finally, **HVXIVG** maps to **SECRET**.

Therefore:

Ciphertext: GSRH RH Z HVXIVG
Plaintext: THIS IS A SECRET

This demonstrates how knowledge of a classical substitution cipher can be used to decrypt an intercepted message.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding g; limited or partial integration	Poor understanding g; unable to integrate concepts
Experiment al Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation on with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation